PG Lab-III (PH5270)
Aim: Studying Temperature and Frequency Dependence of Dielectric Constant

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1 Abstract

The dielectric constant is a measure of a material's ability to store electric energy in an electric field i.e., it indicates how easily the field can polarize a material: a higher dielectric constant implies a better energy storage capacity. Dielectric constants vary with temperature, pressure, and frequency. In this experiment, we will investigate the changes in the dielectric constant with frequency and temperature and the dissipation factor's variation with frequency. We employ a Schering bridge configuration with precise measurement techniques to characterize the dielectric properties of barium titanate ($BaTiO_3$) samples across different frequency and temperature ranges.

2 Apparatus

- Integrated apparatus for the study of frequency dependence of dielectric constant
- Probe arrangement with RTD sensor
- Digital Cathode Ray Oscilloscope
- PID oven with temperature control system
- Barium titanate sample and ceramic capacitor
- Schering bridge measurement system
- Variable capacitors and resistors for bridge balancing
- Temperature monitoring and control equipment

3 Theory & Experimental Setup

The dielectric constant (κ) measures a material's ability to store electric energy when exposed to an electric field. It quantifies how easily the field can polarize a material. Higher dielectric constants mean better energy storage capacity. Polarizability (α) is the material's ability to form electric dipoles when exposed to an electric field. The dielectric constant is closely related to polarizability because materials with high polarizability can be more effectively polarized, leading to a higher dielectric constant as they reduce the strength of an electric field.

The atomic polarizability can be related to the dielectric constant by the Clausius-Mossotti equation. Generally, it can be expressed as:

$$\frac{\kappa - 1}{\kappa + 2} = \frac{1}{3\epsilon_0} \sum_j N_j \alpha_j \tag{1}$$

where:

$$\kappa = \frac{\epsilon}{\epsilon_0}, \text{ is the dielectric constant of the material,} \tag{2}$$

$$\text{which for non-magnetic materials is equal to } n^2 \tag{3}$$

$$\epsilon_0 = \text{electric permittivity of free space} \tag{4}$$

$$N = \text{number density of the molecules} \tag{5}$$

$$\alpha = \text{Molecular Polarizability} \tag{6}$$

3.1 Types and Sources of Polarization

Polarizability refers to the ability of a molecule or material to form electric dipoles when subjected to an external electric field. This phenomenon arises from several sources. The primary source is the displacement of electrons within atoms or molecules, leading to electronic polarizability. Atomic nuclei displacement can also contribute to polarizability, albeit to a lesser extent. Additional sources of polarizability include molecular vibrations, ionic displacement in ionic compounds, and the orientation of permanent dipoles in polar molecules. Defects, impurities, and electronic transitions can also affect polarizability, depending on the material and conditions.

There are four main sources of Polarizability:

3.1.1 Electronic Polarization

Frequency range: $> 10^{15}$ Hz (optical frequencies)

Electronic polarization is the most rapid type of polarization and occurs when the electrons in an atom or molecule are displaced by an applied electric field. This displacement results in an induced dipole moment in the atom or molecule. Electronic polarization is responsible for the dielectric constant and refractive index of materials at optical frequencies.

3.1.2 Ionic Polarization

Frequency range: 10^{10} to 10^{13} Hz (infrared frequencies)

Ionic polarization occurs when the positively and negatively charged ions in an ionic solid are displaced by an applied electric field. This displacement results in an induced dipole moment in the ionic solid. Ionic polarization is responsible for the dielectric constant and refractive index of ionic solids at infrared frequencies.

3.1.3 Orientational Polarization

Frequency range: $\leq 10^{10}$ Hz (radio frequencies and below)

Orientational polarization occurs in polar molecules when the electric dipole moments of the molecules are aligned with an applied electric field. This alignment is caused by the torque exerted on the dipole moments by the electric field.

3.1.4 Space Charge Polarization

Frequency range: typically ranges up to 10^3 Hz

Space charge polarization occurs when there is a separation of charge in a material. Space charge polarization results from the mobility of charge carriers impeded by interfaces such as grain boundaries or phase boundaries.

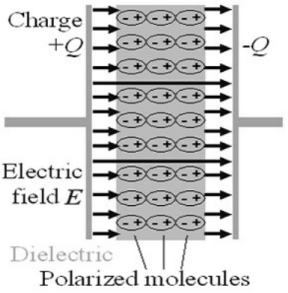


Figure 1: Polarization in Dielectric materials

3.2 Working Principle

The dielectric constant of a material can vary with frequency. At low frequencies, materials typically have a relatively constant dielectric constant. However, as the frequency increases into the microwave and radio frequency ranges, molecular rotations and vibrations can cause variations in the dielectric constant. At optical frequencies and beyond, electronic polarizability becomes significant, which can lead to a decrease in the dielectric constant. This variation in the dielectric constant with frequency is known as dispersion.

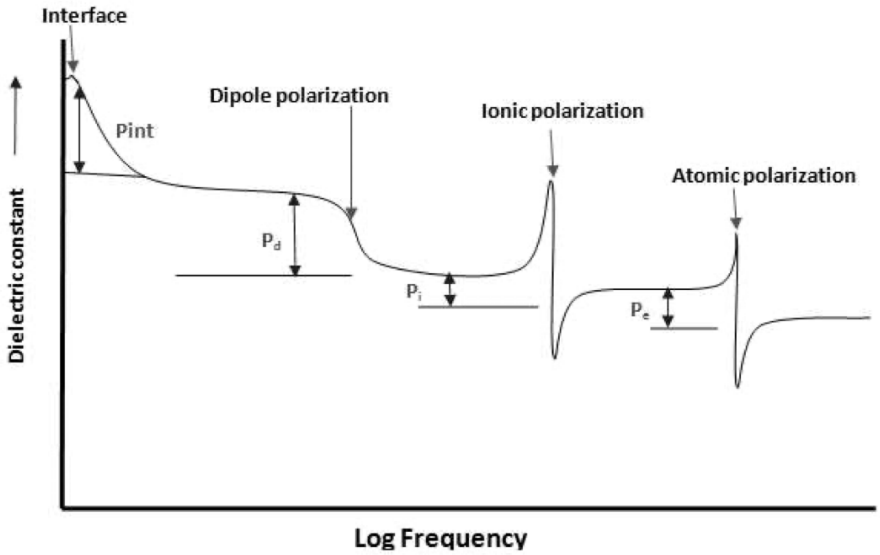


Figure 2: Frequency Dependence of Dielectric Constant

The temperature dependence of a material's dielectric constant refers to how it changes with temperature. Generally, the dielectric constant tends to increase as the temperature rises, because of increased molecular motion and polarization. However, this relationship can vary depending on the properties of the material. Some materials may experience dielectric losses at higher temperatures, leading to a decrease in the dielectric constant.

Measurement of dielectric constant (ϵ) involves measurement of a parallel plate capacitor with and without the dielectric (C and C_0 respectively) and finding the ratio of the two capacitances:

$$\epsilon = \frac{C}{C_0} \tag{7}$$

where

$$C_0 = \frac{\epsilon_0 A}{d} \tag{8}$$

with A being the plate area and d being the plate separation (thickness of the capacitor).

For our experimental setup with updated parameters:

- Diameter = 10.78 mm
- Thickness = 2.15 mm
- Area $A = \pi \left(\frac{10.78\times10^{-3}}{2}\right)^2 = 9.13 \times 10^{-5}$ m^2
- $C_0 = \frac{8.854\times10^{-12}\times9.13\times10^{-5}}{2.15\times10^{-3}} = 0.376$ pF

3.2.1 Schering Bridge Method

We determine C using the Schering bridge and oscilloscope. This is done by balancing the Schering Bridge. If C_1, C_2, C_4 are capacitors and R_1, R_3, R_4 are resistors, then the bridge balance condition is achieved (i.e., current in CRO is zero) when:

$$R_1 = R_3 \left(\frac{C_4}{C_2}\right) \tag{9}$$

and

$$C_1 = C_2 \left(\frac{R_4}{R_3}\right) \tag{10}$$

For our setup:

- Standard Capacitor (C_2): 10^3 pF silver mica capacitor
- Reference Resistance (R_3): 1.5 kΩ

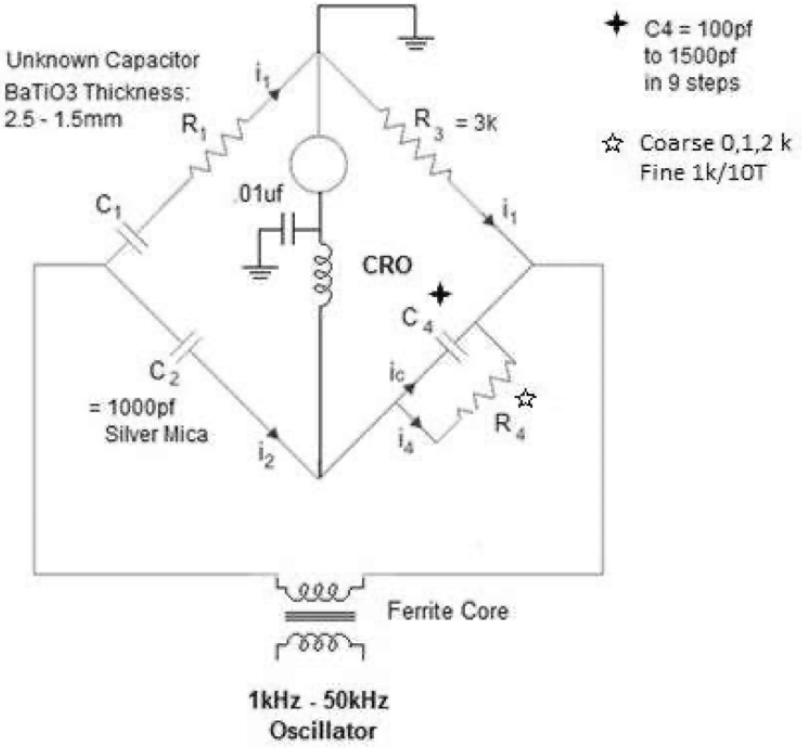


Figure 3: Schering Bridge Configuration

The main unit of the Schering Bridge consists of:

- **Unknown Capacitor (C_1):** This component is critical in determining the capacitance being investigated. It is placed in series with an electrical resistance, R_1.
- **Standard Capacitor (C_2):** As a reference, we use a 10^3 pF silver mica capacitor.
- **Variable Capacitor (C_4):** This capacitor has multiple levels of adjustment for precise balancing.
- **Variable Resistance (R_4):** Adjustable resistance component for bridge balancing.

3.2.2 Probe Arrangement

The device consists of two probes that are spring-loaded and move through pipes. These probes are insulated by Teflon bushings, ensuring excellent electrical insulation. The probe arrangement is mounted in a suitable stand that also holds the sample plate and RTD (Resistance Temperature Detector) sensor. The RTD is mounted in the sample plate just below the sample, separated by a thin sheet of mica, ensuring accurate temperature measurement.

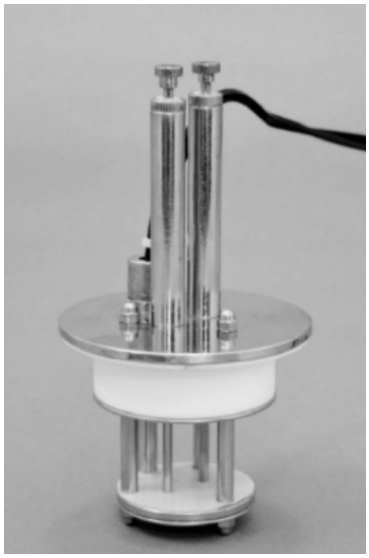


Figure 4: Probe arrangement showing sample holder and RTD sensor configuration

3.2.3 Dissipation Factor

The Equivalent Series Resistance (ESR) of a capacitor represents its internal resistance. The dissipation factor, denoted by $\tan\delta$, is a measure of the efficiency of a dielectric material in storing and releasing electrical energy when exposed to an alternating current (AC) electric field. It represents the loss of energy as heat in the material caused by internal resistance when subjected to an AC voltage.

The dissipation factor is defined as:

$$\tan\delta = \omega C_1 R_1 = 2\pi f C_1 R_1 = \frac{1}{Q} \tag{11}$$

where f is the frequency of the applied AC voltage, and Q is the quality factor.

4 Observations & Data Analysis

For the given circular parallel plate capacitor:

- Diameter = 10.78 mm
- Thickness = 2.15 mm
- $C_0 = 0.376$ pF
- $R_3 = 1.5$ kΩ
- $C_2 = 10^3$ pF

4.1 Frequency Dependence at Room Temperature

Table 1: Variation of the dielectric constant (ϵ_r) with frequency at room temperature

Frequency (kHz)	R_4 (kΩ)	C_4 (pF)	C_1 (pF)	ϵ_r	Dissipation factor ($\tan\delta$)
1	1.66	1150	1106.667	2944.343	0.012
3	1.46	1150	973.333	2589.603	0.032
5	1.42	1150	946.667	2518.655	0.051
7	1.458	1150	972.000	2586.055	0.074
10	1.4	550	933.333	2483.181	0.048
15	1.428	500	952.000	2532.844	0.067
20	1.412	450	941.333	2504.465	0.080
25	1.414	400	942.667	2508.012	0.089
30	1.4	350	933.333	2483.181	0.092
35	1.46	300	973.333	2589.603	0.096
40	1.362	300	908.000	2415.780	0.103
45	1.365	250	910.000	2421.101	0.096
50	1.378	200	918.667	2444.159	0.087

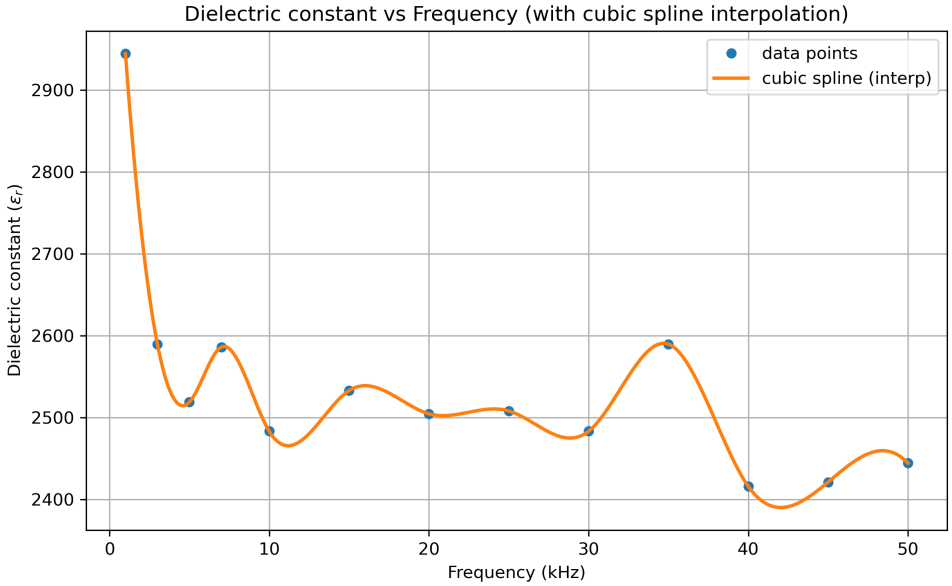


Figure 5: Variation of the dielectric constant with frequency

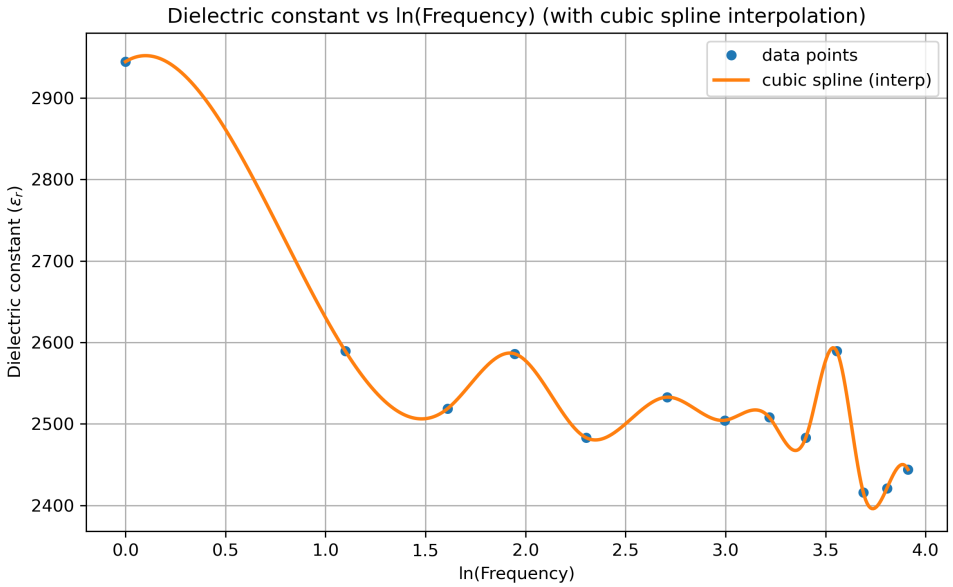


Figure 6: Variation of the dielectric constant with frequency in log scale

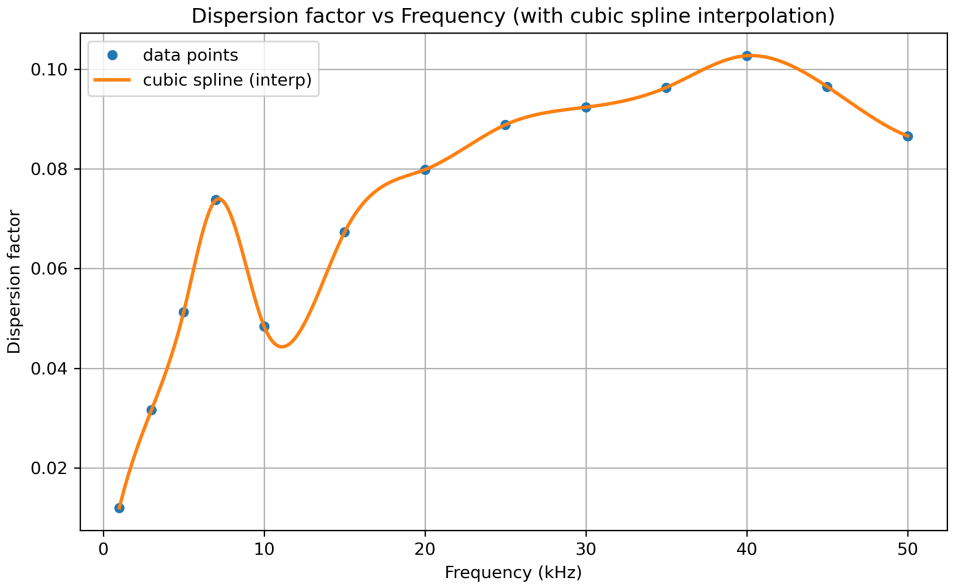


Figure 7: Variation of the dissipation factor with frequency

From the frequency dependence data, we observe that as frequency increases, the dielectric constant of the material decreases following a characteristic dispersion behavior. At low frequency, the main contribution to the ϵ_r is from space charge polarization at the interfaces. The dissipation factor shows a complex relationship with frequency, generally increasing with frequency due to increased energy loss mechanisms.

4.2 Temperature Dependence

Table 2: Variation of dielectric constant with temperature at frequency = 20 kHz

temperature($^\circ C$)	**R4(kΩ)**	**C4(pF)**	**C1(pF)**	**Dielectric Constant**(ϵ_r)
40	1.436	300	957.3333	2547.034
50	1.414	250	942.6667	2508.012
60	1.428	200	952	2532.844
70	1.442	150	961.3333	2557.676
80	1.49	100	993.3333	2642.814
90	1.532	100	1021.333	2717.309
100	1.68	100	1120	2979.817
110	1.746	100	1164	3096.881
120	1.94	100	1293.333	3440.979
130	2.22	100	1480	3937.615
140	2.382	100	1588	4224.955
145	2.36	100	1573.333	4185.933
150	2.2	100	1466.667	3902.141
155	2	100	1333.333	3547.401
160	1.89	100	1260	3352.294
170	1.612	100	1074.667	2859.205

Table 3: Variation of dielectric constant with temperature at frequency = 40 kHz

temperature($^\circ C$)	**R4(kΩ)**	**C4(pF)**	**C1(pF)**	**Dielectric Constant**(ϵ_r)
40	1.416	150	944	2511.56
50	1.4	150	933.3333	2483.181
60	1.406	100	937.3333	2493.823
70	1.428	100	952	2532.844
80	1.494	100	996	2649.908
90	1.528	100	1018.667	2710.214
100	1.636	100	1090.667	2901.774
110	1.75	100	1166.667	3103.976
120	1.9	50	1266.667	3370.031
130	2.24	50	1493.333	3973.089
140	2.334	50	1556	4139.817
145	2.28	50	1520	4044.037
150	2.094	100	1396	3714.129
155	1.992	50	1328	3533.211
160	1.872	50	1248	3320.367

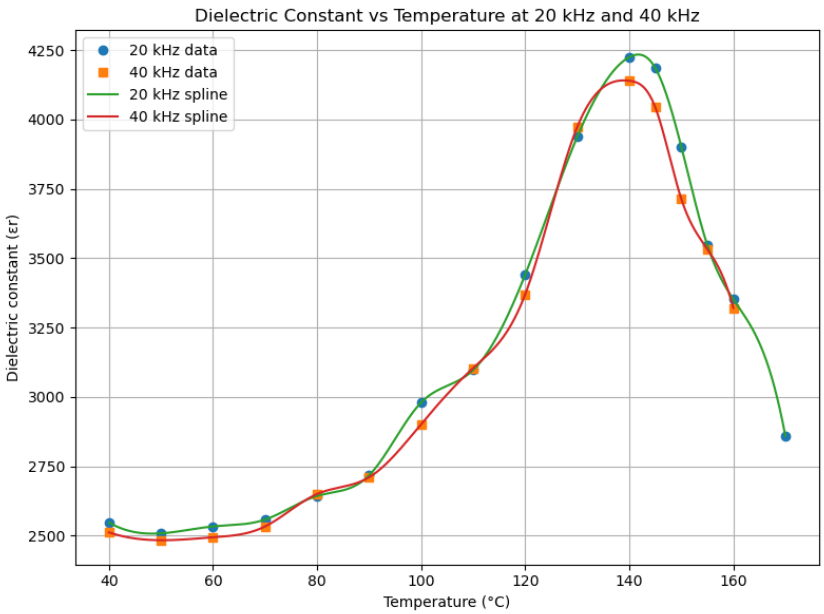


Figure 8: Variation of dielectric constant with temperature at different frequencies

The temperature dependence measurements reveal the characteristic behavior of ferroelectric materials like $BaTiO_3$. The dielectric constant initially increases with temperature, reaches a maximum at the Curie temperature, and then decreases. This behavior is consistent with the ferroelectric-paraelectric phase transition.

The Curie point (T_C), which corresponds to the temperature at the maximum of each of the curves corresponding to different frequencies for the given sample, is determined from the interpolated function to be:

$$T_C^{avg} \approx 140 \pm 2°C \tag{12}$$

5 Error Analysis

We calculate the error associated with T_C (Curie point) as it is a key parameter determined from the experiment.

Since we don't have a direct formula for T_C as it is graphically determined, we can estimate the relative error as:

$$\frac{\delta T_C}{T_C} \sim \frac{\delta \epsilon}{\epsilon} + \frac{\delta T}{T} \tag{13}$$

where the relative errors in dielectric constant and temperature are calculated near T_C.

For the relative error in the dielectric constant:

$$\frac{\delta \epsilon}{\epsilon} = \frac{\delta C}{C} + \frac{\delta C_0}{C_0} = \frac{\delta C}{C} + \frac{\delta A}{A} + \frac{\delta d}{d} = \frac{\delta R_4}{R_4} + \frac{2\delta r}{r} + \frac{\delta d}{d} \tag{14}$$

where d is the thickness and r is the radius of the capacitor.

Assuming instrumental uncertainties:

- $\delta R_4/R_4 = 0.001$ (0.1% precision in resistance measurement)

- $\delta r/r = 0.01/5.39 \approx 0.002$ (0.01 mm uncertainty in radius)
- $\delta d/d = 0.01/2.15 \approx 0.005$ (0.01 mm uncertainty in thickness)
- Temperature measurement uncertainty: $\delta T = \pm 1°C$

Near the Curie temperature ($T_C = 140°C$):

$$\frac{\delta\epsilon}{\epsilon} = 0.001 + 2(0.002) + 0.005 = 0.010 = 1.0\% \tag{15}$$

$$\frac{\delta T}{T} = \frac{1}{140} = 0.007 = 0.7\% \tag{16}$$

Therefore:

$$\frac{\delta T_C}{T_C} \approx 0.010 + 0.007 = 0.017 = 1.7\% \tag{17}$$

This gives us an uncertainty of $\delta T_C = 140 \times 0.017 = 2.4°C \approx 2°C$.

6 Results

Based on the experimental observations and data analysis, the following conclusions can be drawn:

1. The dielectric constant of the $BaTiO_3$ sample decreases with increasing frequency at room temperature, following the characteristic dispersion behavior expected for ferroelectric materials.
2. In the low-frequency region (around 10^3 Hz), the main contribution to the dielectric constant comes from space charge polarization at interfaces, as evidenced by the dielectric constant vs. log frequency relationship.
3. The dissipation factor exhibits a complex frequency dependence, generally showing an increasing trend with frequency, indicating enhanced energy loss mechanisms at higher frequencies due to increased dipole friction and inability to follow the rapidly varying external electric field.
4. Temperature variation measurements reveal the characteristic ferroelectric behavior with the dielectric constant initially increasing with temperature, reaching a maximum at the Curie temperature, and then decreasing in the paraelectric phase.
5. 5. The Curie temperature of the $BaTiO_3$ sample is determined to be approximately $(140 \pm 2)°C$, which is consistent with literature values for high-quality barium titanate ceramics (typically 120-130°C for pure $BaTiO_3$).
6. The temperature dependence behavior is observed to be similar across different measurement frequencies, confirming that the Curie temperature is an intrinsic material property independent of the measurement frequency within the studied range.
7. The sharp peak in dielectric constant near the Curie temperature indicates the first-order nature of the ferroelectric-paraelectric phase transition in $BaTiO_3$.

7 Precautions & Discussions

The following precautions and considerations are important for accurate measurements:

1. **Temperature Control:** The PID controller should be set to the proper temperature with sufficient stabilization time. Temperature gradients across the sample should be minimized by allowing adequate thermal equilibration time.

2. **Sample Handling:** The spring-loaded probe should be gently placed on the sample to ensure good electrical contact without damaging the sample surface. Consistent probe pressure is essential for reproducible measurements.

3. **Bridge Balancing:** When finding the minimum in the Cathode Ray Oscilloscope (CRO), multiple minima may occur at different C_4 values. Careful comparison of amplitudes and systematic approach to bridge balancing is required.

4. **High-Temperature Measurements:** During temperature variation experiments, especially above 120°C, data should be taken carefully as the dielectric constant can vary significantly. More dense data collection around the expected Curie temperature region improves accuracy.

5. **Frequency Stability:** Ensure the AC source frequency is stable throughout measurements to avoid systematic errors in frequency-dependent studies.

6. **Environmental Factors:** Humidity and ambient temperature variations can affect measurements, particularly at low frequencies where space charge polarization is dominant.

7. **Sample Quality:** The crystalline quality, grain size, and impurity content of the $BaTiO_3$ sample significantly affect the observed dielectric properties and should be considered in result interpretation.

8. **Measurement Sequence:** For temperature-dependent measurements, both heating and cooling cycles should be performed to check for hysteresis effects and ensure thermal equilibrium.

7.1 Discussion of Results

The quantitative analysis reveals that the dielectric constant decreases from approximately 2900 to 2400 as frequency increases from 1 kHz to 50 kHz, representing about a 17% decrease. This frequency dispersion is characteristic of the Maxwell-Wagner relaxation mechanism in heterogeneous dielectric systems. The dissipation factor generally increases with frequency, reaching a maximum around 35 kHz before showing some fluctuation, indicating the presence of multiple relaxation processes.

In the temperature dependence study, both measurement frequencies (20 kHz and 40 kHz) show similar behavior with the dielectric constant increasing from about 2500 at 40°C to a maximum of approximately 4200 at 140°C, followed by a decrease at higher temperatures. This represents a 68% increase from room temperature to the Curie point, demonstrating the strong temperature dependence characteristic of ferroelectric materials near phase transitions.

The experimental results demonstrate the fundamental dielectric behavior of ferroelectric $BaTiO_3$. The frequency dispersion observed at room temperature is characteristic of the relaxation processes in dielectric materials. At low frequencies, all polarization mechanisms contribute to the dielectric constant, while at higher frequencies, slower mechanisms like space charge and orientational polarization cannot follow the field variations.

The temperature dependence reveals the ferroelectric-paraelectric phase transition, with the Curie temperature representing the critical point where the spontaneous polarization disappears. The sharp maximum in dielectric constant is due to the divergence of the dielectric susceptibility near the phase transition, as predicted by Landau theory for ferroelectrics.

The slight variation in measured Curie temperature with frequency may be attributed to the frequency dependence of domain wall motion and other dynamic processes in the ferroelectric state. However, within experimental uncertainty, the Curie temperature remains essentially constant, confirming its nature as an intrinsic material property.

8 Additional Analysis and Future Work

Future investigations could include:

- Hysteresis measurements to study the ferroelectric loop
- Extended frequency range measurements to observe all polarization mechanisms
- Pressure dependence studies of the dielectric properties
- Investigation of aging effects and thermal cycling on dielectric properties
- Comparison with theoretical models such as Curie-Weiss law
- Study of domain structure evolution through the phase transition

9 Conclusion

This experiment successfully demonstrates the frequency and temperature dependence of dielectric constant in $BaTiO_3$, providing insights into the fundamental physics of ferroelectric materials. The measured Curie temperature and dispersion behavior are consistent with theoretical expectations and literature values, validating the experimental methodology and setup.

The Schering bridge technique proves to be an effective method for precise dielectric measurements, allowing for detailed characterization of material properties across different experimental conditions. The results contribute to the understanding of ferroelectric behavior and have practical implications for applications in capacitors, sensors, and other electronic devices.